The Sideband Subtraction, Step by Step

How wall–plastic coincidences become clean SiPM-wall MIP peaks

n_TOF EAR2 X17 — run 224404

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analysis and slides prepared with Claude (Anthropic AI assistant)

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The problem the subtraction solves

- In every bunch, wall hits and plastic hits also line up *by chance* (pile-up), not just from real through-going particles.
- These accidental pairs fall inside the coincidence time window too, and they pollute the amplitude spectrum at low amplitude.
- **Idea:** the accidentals are *flat in time* — so measure their amplitude shape in an off-peak **sideband**, scale it to the width of the signal window, and subtract.

signal window = MIP + accidentals

sideband = accidentals only

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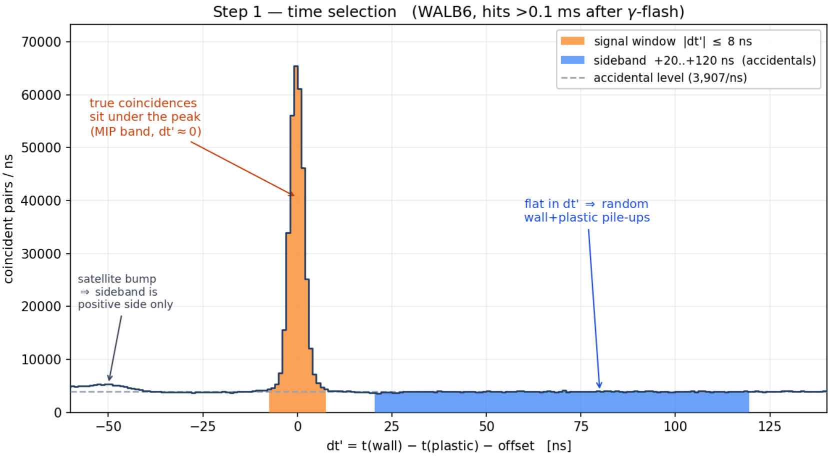
true coincidences

$$= \text{signal} - \frac{16\,\text{ns}}{100\,\text{ns}} \times \text{sideband}$$

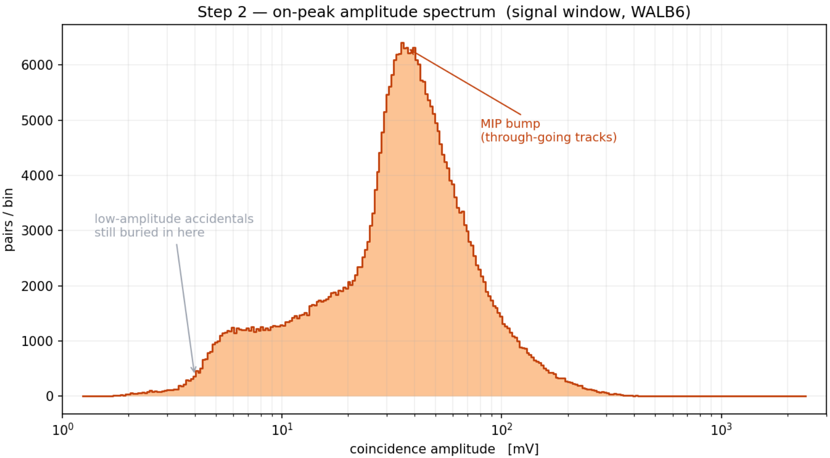
All plots below: one representative channel **WALB6**, high-purity sample (post-recovery bunches, plastic hits $>$0.1 ms after the γ-flash).

Step 1 — the two time windows



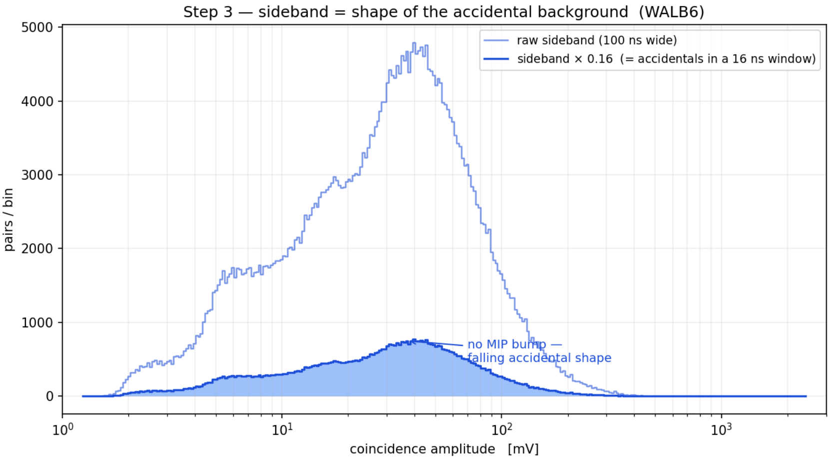
After subtracting the per-channel offset, real pairs peak at $\mathrm{dt}' = 0$. Signal: $|\mathrm{dt}'| \leq 8$ ns. Sideband: $+20$ to $+120$ ns — positive side only, to dodge the -50 ns satellite bump. The sideband floor *is* the accidental rate.

Step 2 — amplitude spectrum inside the signal window



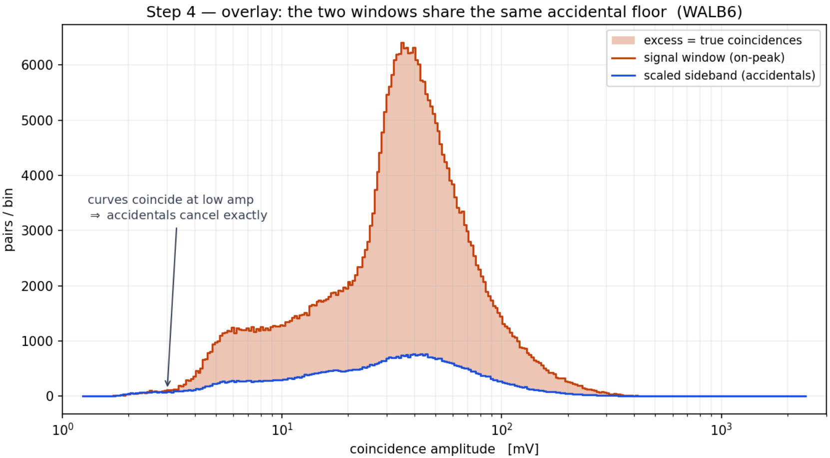
This is what you get with a plain time cut: a clear MIP bump near 35 mV, *but* sitting on a large low-amplitude tail of accidental pairs. Not yet a calibration-grade peak.

Step 3 — the sideband is the accidental shape



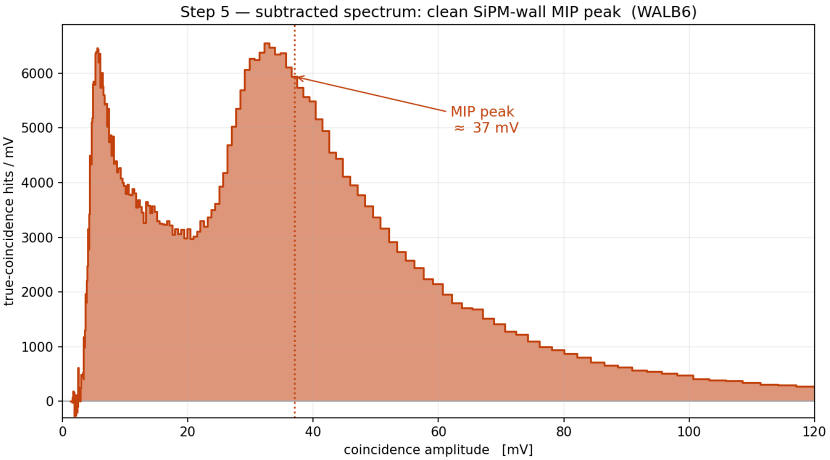
Same pairs, but from the off-peak window: **no MIP bump** — just the falling accidental amplitude shape. Scaling by 16/100 converts the 100 ns-wide sideband into the accidental content expected inside the 16 ns-wide signal window.

Step 4 — overlay: same accidental floor



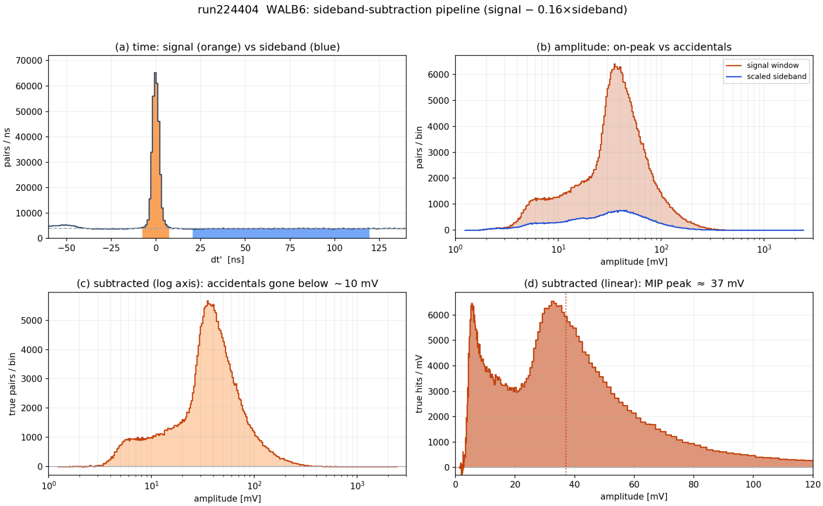
The two curves lie on top of each other at low amplitude — the accidentals are identical — and separate only where the real MIP population lives. The shaded gap is the true-coincidence excess.

Step 5 — subtract ⇒ clean MIP peak

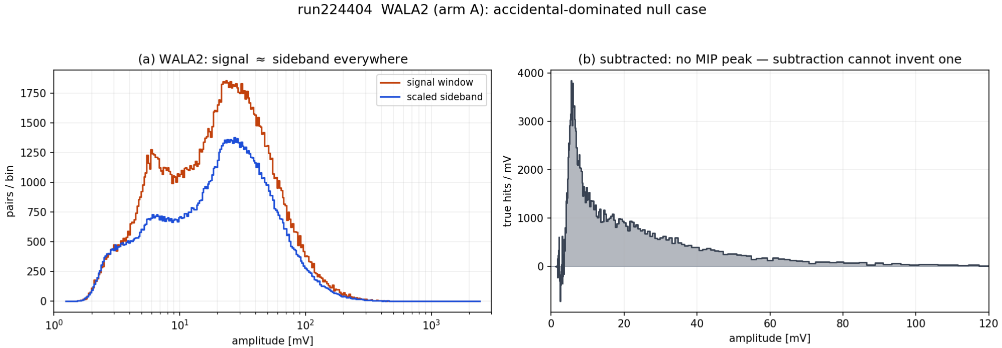


signal − 0.16 × sideband: the accidental tail cancels and a clean SiPM-wall MIP peak at ≈ 37 mV remains — this is the per-channel calibration landmark. (The narrow ∼5 mV feature is a real low-amplitude coincidence population, present in all arms.)

The whole pipeline on one slide (WALB6)

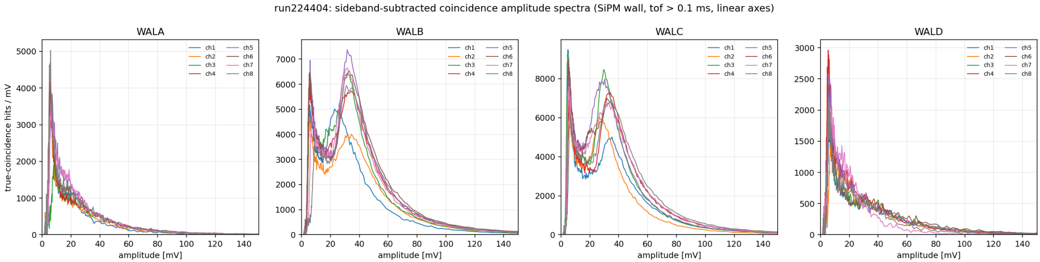


Sanity check: subtraction cannot manufacture a peak



Arm-A channel WALA2: here the signal window is accidental-dominated, so signal and scaled sideband have the *same* shape. After subtraction the residual just falls off — **no fake MIP peak**. The method only reveals a peak where a real one exists (arms B/C), never invents one.

Same recipe, all 16 B/C channels



Each curve is one wall channel's subtracted spectrum. **B/C: clean MIP peaks 29–39 mV on all 16 channels** ⇒ per-channel gain landmark. A/D: accidental-dominated (null case above), MIP cause still open.

Summary

- The “nice” MIP peaks are **not** the raw in-time spectrum — they are signal $-$ 0.16 $\times$ sideband.
- Signal $= |\mathrm{dt}'| \leq 8$ ns (MIP $+$ accidentals); sideband $= +20$ to $+120$ ns (accidentals only), scaled by the window-width ratio $16/100 = 0.16$.
- The subtraction removes the low-amplitude accidental tail exactly, leaving a calibration-grade SiPM-wall MIP peak wherever real through-going tracks exist.
- Verified null: where there is no MIP population, the subtraction returns no peak.